

Biobanking and Genomic Research



HAMILTON[®]

Automation for Biobanking and Genomic Research

Hamilton Provides Complete Solutions for Your Laboratory Workflows



Pre-Processing

- E** **easyBlood STARlet** for automated blood fractionation (see page 7).

Storage

- K** **ASM™** for automated high-throughput sample management of tubes, plates and vials (see page 9).
- L** **ASM and Microlab® STAR** for automated liquid handling integration (see page 9).
- N** **SAM™** for automated low-capacity storage of tubes, plates and vials (see page 10).
- O** **BiOS™ M and L** for medium- to large-capacity automated storage for sensitive samples in tubes, vials and plates (see page 10).
- P** **BiOS XL** for extra-large-capacity automated storage for sensitive samples in tubes, vials and plates (see page 11).

Post-Processing

- R** **Microlab NIMBUS® Select** automated electrophoresis for both size selection and analytical purposes (see page 12).
- S** **Microlab NIMBUS PCR** for automated PCR assay setup and sample normalization (see page 13).
- T** **Microlab NEB® NGS Workstation** for library preparation for Illumina® Sequencing with NEB® reagents (see page 12).
- U** **Microlab PCR Setup STARlet** for automated PCR assay setup (see page 13).
- V** **Illumina® NGS STAR Workstation** for automated Illumina® NGS sample preparation (see page 11).

Pre- and Post-Processing

- A** **LabElite™ DeCapper™** for automated decapping/capping of screw cap tubes and cryovials (see page 3).
- B** **LabElite I.D. Capper™** for automated decapping/capping and barcode reading in one device (see page 4).
- C** **LabElite I.D. Reader™** for automated barcode reading of samples (see page 3).
- D** **Genomic STARlet** for automated nucleic acid extraction (see page 8).
- F** **Aliquoting and Reformatting STAR** for aliquoting of biological samples and reformatting samples into automation-friendly labware (see page 6).
- G** **Microlab STAR** for automated liquid handling for genomics, forensics, proteomics, and cellomics applications (see page 5).
- H** **Microlab AutoLys STAR** for performing automated sample lysis and extraction (see page 7).
- I** **chemagic STAR** for automated nucleic acid extraction of human samples (see page 8).
- J** **Microlab NIMBUS** for automating a single or select set of liquid handling routines (see page 5).
- M** **Normalization STAR** for DNA normalization (see page 6).
- Q** **Rack Runner** for integration of sample storage systems to several devices (see page 4).



Automated Barcode Reading

C LabElite I.D. Reader



Purpose: Automated barcode reading of 2-D barcoded tubes and 1-D barcoded racks.

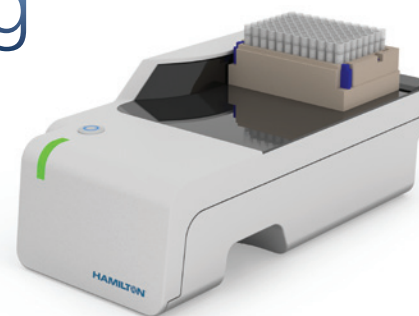
Applications

The high-speed I.D. Reader automatically decodes 2-D barcoded tubes stored in SBS racks from all major brands, supporting complete sample tracking in your laboratory. Intelligent features like automatic rack type detection and integrated 1-D barcode reading simplify your workflow and set a new standard for tube rack readers.

Reference Material

LabElite Line Brochure (Lit. No. HST-BR0005)

LabElite I.D. Reader Flyer (Lit. No. HST-F0007)



Key Features

- ▶ Process a 96-tube rack in less than 3 seconds and a 384-tube rack in 5 seconds
- ▶ Optional 1-D barcode reading for racks
- ▶ Automatically detects the tube rack type to use optimal settings
- ▶ Highly-configurable output options for smooth integration into LIMS or databases

Automated Decapping

A LabElite DeCapper



Purpose: Automated decapping/capping of screw cap tubes and cryovials.

Applications

The next-generation DeCapper is an innovative, easy-to-use device that provides automated decapping/capping of tubes in both 48- and 96-tube racks, with internal or external threads. The DeCapper processes racks in both portrait and landscape orientation, and can decap both 48- and 96-tube racks on a single device by simply exchanging the decapping head (patent pending).

Reference Material

LabElite Line Brochure (Lit. No. HST-BR0005)

LabElite DeCapper Flyer (Lit. No. HST-F0006)



Key Features

- ▶ Process a full rack of tubes or row of tubes in portrait or landscape format
- ▶ Can be operated as an integrated and standalone device
- ▶ Decaps one row at a time to limit risk of exposure or contamination from environment



Automated Decapping & Barcode Reading

B LabElite I.D. Capper



Purpose: Automated decapping/capping and barcode reading in one device.

Applications

The all-in-one I.D. Capper enables labs to combine decapping/capping and high-speed barcode reading within one device without any additional user interaction. This is the first device in the market that maintains an audit trail for decapped samples—identifying the samples and recording the time of opening—providing a new level of sample tracking within your laboratory.

Reference Material

LabElite Line Brochure (Lit. No. HST-BR0005)

LabElite I.D. Capper Flyer (Lit. No. HST-F0008)



Key Features

- ▶ Tracks samples at the point of opening for processing
- ▶ Decap/cap both 48- and 96-tube racks in one device
- ▶ Decode 2-D barcodes on a 96-tube rack in less than 3 seconds and a 384-tube rack in 5 seconds

Liquid Handling Integration

Q Rack Runner



Purpose: Robot used to integrate sample storage systems to several devices.

Applications

The Rack Runner is a fast and reliable robot used to integrate our automated storage systems to several devices including Microlab STAR Line workstations, additional storage systems, DeCappers, or third-party equipment such as hotels, incubators and plate readers.

Reference Material

Automated Sample Management Product Catalog (Lit. No. HST-BR0002-01)

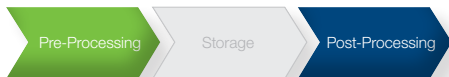


Key Features

- ▶ Compact footprint
- ▶ Linear drive technology
- ▶ Absolute encoders eliminate slow and risky initialization movements and collisions with peripheral devices

Liquid Handling

J Microlab NIMBUS



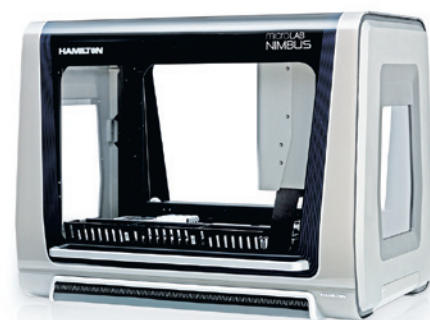
Purpose: A small-scale liquid handler designed for space and budget-conscious labs.

Applications

The Microlab NIMBUS is a compact, multi-channel automated liquid handler, offering speed, flexibility, ease-of-use and superior pipetting performance. NIMBUS is ideally suited for automating a single or select set of liquid handling routines for applications such as DNA/RNA extraction and purification, microarray sample preparation, PCR setup and purification, liquid-liquid extraction, post-PCR cleanup, and genomics assays.

Reference Material

Microlab NIMBUS Brochure (Lit. No. BR0001)



Key Features

- ▶ Small footprint
- ▶ Entry-level pipettor
- ▶ Flexible deck layout
- ▶ Broad range of modular accessories for quick and easy reconfiguration

Liquid Handling

G Microlab STAR



Purpose: Performing liquid handling applications in diverse application areas such as life sciences, biotech and pharmaceutical industries.

Applications

Microlab STAR is an automated pipetting workstation that can be used for several medium and high-throughput liquid handling applications in a wide variety of areas such as genomics, forensics, proteomics and cellomics. The powerful VENUS software package allows flexible and individual programming of all kinds of workflows. A set of sophisticated monitoring features provide full monitoring of each liquid transfer. A robotic plate handler allows integration of third-party devices such as readers, hotels, centrifuges, incubators and sealers.

Reference Material

Microlab STAR Line Brochure (Lit. No. BR0010)



Key Features

- ▶ Compressed O-Ring Expansion (CO-RE) technology
- ▶ Total Aspiration and Dispense Monitoring (TADM) technology
- ▶ Monitored Air Displacement (MAD) technology
- ▶ Anti-Droplet Control (ADC)
- ▶ Scalable in size
- ▶ Freely programmable



Aliquoting and Reformatting

F Microlab STAR



Purpose: Aliquoting of primary samples and reformatting of archived samples.

Applications

This customized STAR solution provides automated aliquoting from all kinds of primary containers into final storage tubes or plates. It is being used for urine, plasma, serum, whole blood and other liquids. As a reformatting station, the system transfers liquids from archived sample tubes into automation-friendly 2-D barcoded storage tubes or 1-D barcoded storage plates. The full audit trail can be provided when combined with a 1-D barcode reader with autoloader function and the on-deck 2-D barcode reader.

Reference Material

Microlab STAR Line Brochure (Lit. No. BR0010)



Key Features

- ▶ Supports wide range of labware
- ▶ Fully monitored liquid transfer with Anti-Droplet Control (ADC)
- ▶ Total Aspiration and Dispense monitoring (TADM)
- ▶ Monitored Air Displacement (MAD) technology
- ▶ Freely programmable to easily adapt to new tube formats, liquids and protocols

Normalization

M Microlab STAR



Purpose: Normalization of DNA concentration.

Applications

This customized STAR provides normalization of DNA samples before storage or delivery. This customized solution integrates a DNA concentration reader that is being used to read the DNA concentration of the original sample. The system automatically dilutes the primary sample to equalize DNA concentration across all samples.

Reference Material

Microlab STAR Line Brochure (Lit. No. BR0010)



Key Features

- ▶ Freely programmable to be used for other applications
- ▶ Fully automated process available when using integrated 1-D or 2-D barcode readers and barcoded labware
- ▶ Full monitoring of all liquid transfers

Blood Fractionation

E easyBlood STARlet



Purpose: Blood fractionation based on image analysis of sample tubes. Plasma aliquoting, buffy coat pipetting and red blood cell distribution in one instrument.

Applications

The easyBlood STARlet splits up a blood sample into the three fractions—plasma, buffy coat and red blood cells—directly out of the primary tube. A camera is used to determine the location of blood layers and volumes. After removing plasma, a dedicated tip is used for buffy coat pipetting to ensure a high recovery rate of white blood cells. As a last step, red blood cells can be collected. Additionally, plasma and serum from gel-plug blood collection tubes can be pipetted into destination containers.

Reference Material

easyBlood STAR Brochure (Lit. No. B-1203-03—00/2011)

easyBlood Flyer (Lit. No. F-1106-01—03/2012)

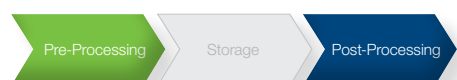


Key Features

- ▶ Buffy coat isolation
- ▶ Plasma and red blood cell aliquoting
- ▶ Compatible with a wide range of primary tubes
- ▶ Customizable detection parameters to maximize imaging reliability
- ▶ 2-D barcode reader integration

Sample Preparation

H Microlab AutoLys STAR Workstation



Purpose: Performing automated sample lysis and extraction.

Applications

The Microlab AutoLys STAR Workstation provides a fully automated solution for lysis and extraction from a variety of sample types including blood spots, tissues, FFPE samples, swabs, etc. Combination of the AutoLys channels and AutoLys tubes offer complete process safety and walk-away automation. This workstation eliminates the sample preparation bottlenecks such as tedious handling of spin columns, manual pipetting and centrifugation.

Reference Material

Microlab AutoLys STAR Flyer (Lit. No. F0003)



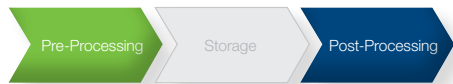
Key Features

- ▶ Labor savings—eliminates tedious spin-column handling and manual pipetting steps
- ▶ Time savings—performs sample lysis completely unattended
- ▶ Enhanced sample integrity—minimizes the risk of contamination and offers chain of custody



Nucleic Acid Extraction

I chemagic STAR



Purpose: Combining chemagen's nucleic acid extraction technology with the Hamilton STAR.



Applications

The chemagic STAR combines chemagen's nucleic acid extraction technology based on magnetic beads with the technology of the Hamilton STAR. The system can extract genomic DNA, total RNA or pathogenic NA out of human samples from 20 μ L up to 10 mL. A broad range of human sample material (e.g., blood, plasma, buffy coat, saliva, cells, tissue, buccal swabs) can be processed with the chemagic STAR. The extracted nucleic acids are ready to use in the fields of HLA-typing, human genetics, pathogen diagnostics and virus screening.

Reference Material

chemagic STAR Brochure (Lit. No. B-1001-02_03)

www.chemagicstar.com

Key Features

- ▶ Most flexible NA extraction platform
- ▶ Based on chemagen extraction technology
- ▶ Broad coverage of human sample material
- ▶ Processes large volume range (20 μ L–10 mL)
- ▶ Thirty-four “ready-to-start” automation protocols

Nucleic Acid Extraction

D Genomic STARlet



Purpose: Performing nucleic acid extraction using silica filter plate-based technology from MACHERY-NAGEL.

Applications

The Genomic STARlet isolates nucleic acids from diversified biological samples such as human blood, cells, and tissue; DNA from plants, plasmids, total RNA from human tissue/cells, viral NA from human plasma or purified PCR products.

Reference Material

Automated Genomic DNA Isolation from Mouse Tails on the Genomic STARlet App Note (Lit. No. AN-1212-03)

Automated Genomic DNA Isolation from Blood App Note (Lit. No. AN-1212-01)

Automated Plasmid Purification on the Genomic STARlet App Note (Lit. No. AN-1212-02)

Automated Total RNA Isolation from Cells and Tissues on the Genomic STARlet App Note (Lit. No. AN-1301-01)



Key Features

- ▶ Performs NA extraction on diversified sample range in one system
- ▶ 8-in-1 system: up to eight verified protocols
- ▶ MACHERY-NAGEL extraction technology
- ▶ 96 samples processed in 60 to 144 minutes (dependent on kit)

Liquid Handling Integration

L ASM & STAR Integration



Purpose: Integrated liquid handling and sample storage to provide a comprehensive sample processing center.

Applications

The integrated system can perform numerous applications including decapping, daughter plate production, and a myriad of pre- and post-processing applications.

Reference Material

Automated Sample Management Product Catalog (Lit. No. HST-BR0002-01)

Hamilton DNA Biobank Application Note (Lit. No. HST-AN0001)

Automated Biobanking at Estonian Genome Center Application Solution (Lit. No. HST-AS0001)



Key Features

- ▶ Automatically retrieve samples from storage for processing
- ▶ Flexible configuration for liquid handling applications
- ▶ Automated decapping and capping with an integrated DeCapper
- ▶ Easily integrate readers, sealers, shakers and other 3rd party devices for expanded applications

-20°C Sample Storage

K ASM



Purpose: High-throughput sample storage and retrieval for tubes, vials and plates.

Applications

The ASM provides automated storage and retrieval of samples in 2-D barcoded tubes or 1-D barcoded plates at temperatures down to -20°C. It is used as a storage and distribution system for DNA, si-RNA, compounds, biological samples and chemical materials.

Reference Material

Automated Sample Management Product Catalog (Lit. No. HST-BR0002-01)

Hamilton DNA Biobank Application Note (Lit. No. HST-AN0001)

Automated Biobanking at Estonian Genome Center Application Solution (Lit. No. HST-AS0001)



Key Features

- ▶ Storage of up to 207,360 tubes or 2,160 plates per module
- ▶ High-speed picking rates of 300 to 1,000 tubes per hour
- ▶ Integrates to Microlab STAR for a fully automated workflow



-20°C and -80°C Sample Storage

N SAM



Purpose: Low-capacity automated biobank for tubes, vials and plates.

Applications

The SAM provides fully automated and secure storage and retrieval of biological and chemical samples. Up to 60,000 samples or 803 plates are stored at temperatures down to -80°C. The small footprint easily fits into any laboratory. The integrated tube picking device allows retrieval of selected tubes identified by 1-D or 2-D barcodes. The system software provides temperature monitoring, access control to samples, a full audit trail for each sample and can be integrated into existing LIMS systems.

Reference Material

Automated Sample Management Product Catalog (Lit. No. HST-BR0002-01)

-80°C Sample Storage

O BIOS M and L



Purpose: Medium- to large-capacity automated biobank for tubes, vials and plates.

Applications

BIOS M and L provide secure automated -80°C storage and retrieval for sensitive biological samples in a compact footprint. The system provides the upmost temperature stability even during sample picking and sample loading/unloading. BIOS M is designed to fit into existing laboratories with limited space, and is intended for forensic applications and medium-sized clinical repositories.

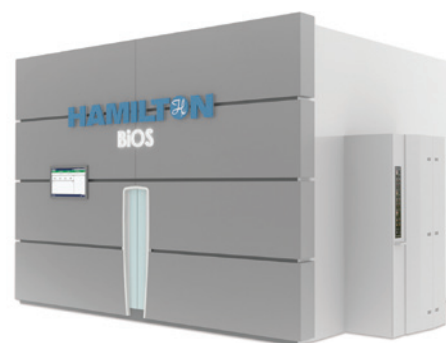
Reference Material

Automated Sample Management Product Catalog (Lit. No. HST-BR0002-01)



Key Features

- ▶ Available as a -20°C or -80°C storage system
- ▶ Tube picking at negative temperatures avoids freeze-thaw cycles for samples while sorting
- ▶ Integrated 1-D and 2-D barcode reading
- ▶ Supports full chain-of-custody



Key Features

- ▶ Small footprint
- ▶ Energy efficient
- ▶ -80°C tube picking to minimize warming effects on samples
- ▶ Sample capacity from 100,000 to 1 million samples

-80°C Sample Storage

P BIOS XL



Purpose: Large- to extra-large-capacity automated biobank for tubes, vials and plates.

Applications

BIOS XL is designed for very large biological sample collections that require secure automated -80°C storage and retrieval. The system provides the upmost temperature stability even during sample picking and sample loading/unloading. It is intended for large clinical biorepositories and population-based biobanks.

Reference Material

Automated Sample Management Product Catalog (Lit. No. HST-BR0002-01)

BIOS Flyer (Lit. No. HST-F0003)



Key Features

- ▶ -80°C tube picking to minimize warming effects on samples
- ▶ Sample capacity from 250,000 to more than 10 million samples
- ▶ Extendable in the field by adding capacity without removing samples
- ▶ Reliable operation by separating storage at -80°C from the automation

Next-Generation Sequencing

V Illumina® NGS STAR Workstation



Purpose: Automated Illumina® NGS sample preparation.

Applications

Automating sequencing library preparation minimizes errors, reduces hands-on time, and enables higher throughput. Illumina's® TruSeq® and Nextera kits are designed with master-mixed reagents and simplified sample prep protocols to facilitate automation. The Illumina® NGS STAR Workstation is designed for throughput of up to 96 samples and provides "Illumina® Qualified" methods for the TruSeq® Stranded Total RNA and mRNA protocols.

Reference Material

Illumina® NGS Sample Prep Automation from Hamilton Flyer (Lit. No. F0029)

Automated Illumina® NGS Sample Preparation: TruSeq® Stranded mRNA on the Hamilton Microlab STAR (Lit. No. L50154)

Next Generation Sequencing Method for Illumina® TruSeq® DNA Sample Preparation Protocol on the Hamilton STAR (Lit. No. L50149)



Key Features

- ▶ "Illumina® Qualified" methods for RNA library preparation
- ▶ Fully automated workflow for processing up to 96 samples per run
- ▶ Integrated process safety through barcode tracking of Illumina® kit components



Next-Generation Sequencing

T Microlab NEB® NGS Workstation



Purpose: Library preparation for Illumina® Sequencing with NEB® reagents.

Applications

The NEB® NGS Workstation is a compact, flexible, open and future-proof workstation for library preparation, all common sequencing platforms and any library prep chemistry. It is well suited for any throughput from 8 to 48 samples and reduces time consuming manual pipetting steps and sample-to-sample variability. Validated methods are available for the NEBNext® Ultra™ DNA and RNA library prep kits for Illumina® sequencers from New England Biolabs (NEB®).

Reference Material

NGS STARlet Flyer (Lit. No. F-1310-01)

Automated Library Preparation using NEBNext® Ultra™ DNA Library Prep Kit Application Note (Lit. No. AN-1401-01)



Key Features

- ▶ Versatile workstation for library preparation
- ▶ Validated methods for NEBNext® Ultra™ kits from New England Biolabs®
- ▶ Available as a STARlet or STAR configuration

Next-Generation Sequencing

R Microlab NIMBUS Select Workstation



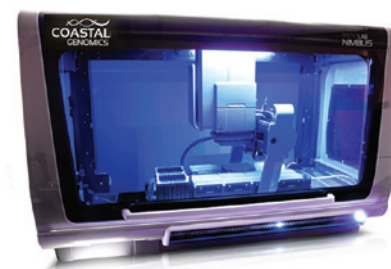
Purpose: Automated gel size selection and analytics.

Applications

The NIMBUS Select Workstation combines Coastal Genomics' Ranger Technology for analytical and size selection processes with the Microlab NIMBUS 96-channel personal pipetting system. Size Selection Mode isolates up to 96 user-defined targets in a sequencing-compatible buffer. Pre- and post- size selection traces are generated for all samples. In Analytics Mode, up to 192 samples can be characterized in under an hour, and 384 sample capacity for genotyping purposes is under development. The NIMBUS Select can also perform general liquid handling tasks.

Reference Material

Microlab NIMBUS DNA Size Selection System Flyer (Lit. No. F0028)

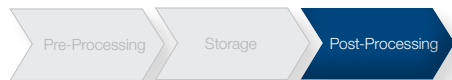


Key Features

- ▶ High recovery size selection and accurate analytics for up to 96 samples
- ▶ Low cost, high throughput option for electropherogram characterization of samples
- ▶ Ideal for next-generation sequencing and gene synthesis applications

PCR Setup

U Microlab PCR Setup STARlet



Purpose: Automated PCR assay setup and sample normalization.

Applications

The PCR Setup STARlet provides setup of endpoint PCR, real-time PCR, qPCR or multiplex PCR for applications such as genotyping, mutagenesis, gene sequencing (Sanger) or pathogen detection.

Reference Material

PCR Setup STARlet Flyer (Lit. No. F-1210-02)

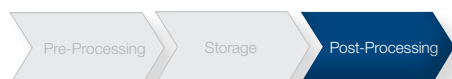


Key Features

- ▶ PCR setup and sample normalization in one software
- ▶ Multiplexing, master mix preparation and serial dilution features
- ▶ Free plate design of 96/384 PCR plates and different formats possible
- ▶ Saves reagents and time resources

PCR Setup

S Microlab NIMBUS PCR Workstation



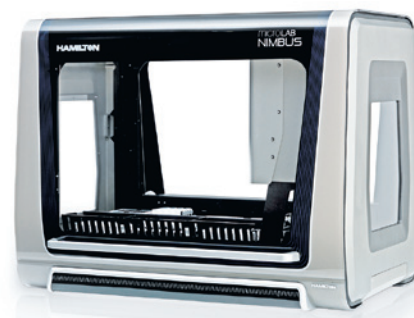
Purpose: Automated PCR assay setup and sample normalization.

Applications

The NIMBUS PCR Workstation provides setup of endpoint PCR, real-time PCR, qPCR or multiplex PCR. The workstation is suitable for a wide range of applications such as gene expression, genotyping, sequencing or pathogen detection. The compact instrument will fit most existing benchtops. It offers speed, flexibility, ease-of-use and superior pipetting performance.

Reference Material

NIMBUS PCR Workstation Flyer (Lit. No. F0020)



Key Features

- ▶ Up to 6 master mixes
- ▶ Tip size selection
- ▶ A liquid class database
- ▶ Up to 192 PCR reactions
- ▶ Sample tracking



About Hamilton Company

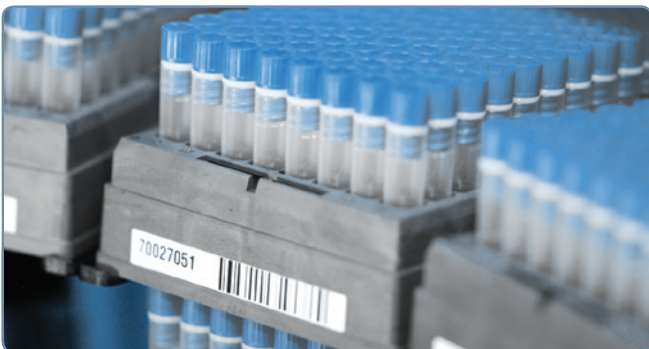
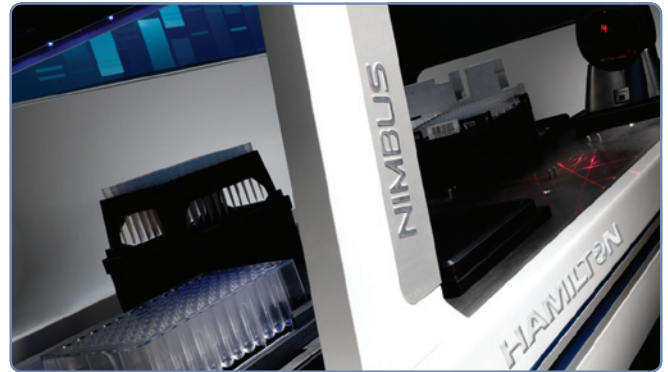
Hamilton Company is a global enterprise with headquarters in Reno, Nevada, Hopkinton, Massachusetts, and Bonaduz, Switzerland and subsidiary offices throughout the world.

We are an industry leader in the design and manufacture of liquid handling, process analytics, robotics and automated storage solutions. For more than 50 years, Hamilton has been satisfying customer needs by combining quality materials with skilled workmanship to ensure the highest level of performance. Hamilton's lifelong commitment to precision and quality has earned us global ISO 9001 Certification.



Founded on the technology of analytical Microliter™ and Gastight® syringes, Hamilton has a broad offering of laboratory products including manual and semi-automated precision fluid measuring instruments, chromatography products, process sensors, laboratory electrodes, pipettes and more. Top innovations from these lines include Arc™ pH, DO and conductivity intelligent sensors, the BioLevigator™ and Microlab 600 diluters/dispensers.

A pioneer in liquid handling equipment and laboratory automation technology, Hamilton Robotics is known for advancing life science and biotechnology industries through reliability, performance and flexibility. Hamilton continues to be the market leader with patented technologies such as Compression-induced O-Ring Expansion (CO-RE), Total Aspiration and Dispensing Monitoring (TADM) and Anti Droplet Control (ADC). Hamilton's feature platform, Microlab STAR, is the highest selling automated pipetting platform for several years running, while Microlab NIMBUS is the first in class of compact, high-speed personalized pipetting workstations.



The Hamilton Storage Technologies division offers comprehensive ultra-low temperature automated sample management systems for a broad array of life science processes. Its line of biobanking, compound storage and consumable solutions consists of products such as BiOS, SAM and ASM, designed for sample integrity, flexibility, and reliability.

Hamilton Company is focused on blending invention and accuracy to deliver customers unparalleled products.



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